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Unit 3: Computer Networks

Complete Exam Question Bank — MCQs, Short Questions & Long Questions

Section A: Multiple Choice Questions

Each question has one correct answer. The correct answer is in bold.

Topic: Network as a System

1. A computer network is best defined as:

- a) A single computer with multiple monitors
 - b) **A system of linked devices that can exchange data and operate together**
 - c) A collection of software programs on one computer
 - d) A storage device that holds large amounts of data
-

2. The primary objective of a computer network is:

- a) To increase the processing speed of a single computer
 - b) To enhance graphic capabilities of devices
 - c) **To enable resource sharing, data communication, and collaboration**
 - d) To improve software development tools
-

3. Which of the following is the largest network in the world?

- a) LAN
 - b) MAN
 - c) WAN
 - d) **The Internet**
-

4. The Internet is often called the:

- a) Global computing system
 - b) World Wide Web
 - c) **Network of networks**
 - d) Universal server
-

5. Which of the following is an example of resource sharing in a computer network?

- a) One computer downloading music
 - b) A student watching a video offline
 - c) **Multiple computers sharing a single printer**
 - d) A computer running two programs at once
-

6. Which component of a network refers to any device connected to it, such as a computer or printer?

- a) Link
 - b) Router
 - c) Switch
 - d) **Node**
-

7. Which component of a network refers to the connection between nodes?

- a) Node
 - b) Router
 - c) **Link**
 - d) Protocol
-

8. Who invented the World Wide Web (WWW)?

- a) Bill Gates
 - b) Steve Jobs
 - c) Alan Turing
 - d) **Tim Berners-Lee**
-

9. In which year was the World Wide Web invented?

- a) 1975
 - b) 1983
 - c) **1989**
 - d) 1995
-

10. Which of the following is an example of connectivity and collaboration through a network?

- a) Printing a document on a local printer
 - b) Saving a file to the desktop
 - c) **A team editing a shared document in real time using Google Drive**
 - d) Installing software from a CD
-

Topic: Data Communication

11. Data communication is defined as:

- a) Storing data on a hard drive
 - b) Processing data using the CPU
 - c) **The exchange of data between a sender and a receiver through a communication medium**
 - d) Converting analog signals to digital signals
-

12. How many basic components does data communication have?

- a) Three
 - b) Four
 - c) **Five**
 - d) Seven
-

13. Which component of data communication is the device that sends data?

- a) Receiver
 - b) Medium
 - c) Protocol
 - d) **Sender**
-

14. Which component of data communication is the actual data being sent?

- a) Protocol
 - b) Sender
 - c) Medium
 - d) **Message**
-

15. A protocol in data communication is best described as:

- a) The physical cable connecting two devices
 - b) The device that receives data
 - c) **A set of rules that govern data communication**
 - d) The speed at which data travels
-

16. Which of the following is an example of a protocol used in web communication?

- a) DNS
 - b) **HTTP**
 - c) MAC
 - d) RAM
-

17. Which component of data communication refers to the physical or wireless path through which data travels?

- a) Protocol
 - b) Message
 - c) Sender
 - d) **Medium**
-

18. An Ethernet cable is an example of which component of data communication?

- a) Sender
 - b) Protocol
 - c) Message
 - d) **Medium**
-

19. When a computer sends an email, it is acting as the:

- a) Receiver
 - b) Medium
 - c) Protocol
 - d) **Sender**
-

20. When a smartphone receives an email, it is acting as the:

- a) Sender
 - b) **Receiver**
 - c) Medium
 - d) Protocol
-

Topic: Networking Devices

21. Which networking device connects multiple devices within a single network and uses MAC addresses to forward data?

- a) Router
 - b) Access Point
 - c) Hub
 - d) **Switch**
-

22. A switch operates at which layer of the OSI model?

- a) Layer 1 — Physical Layer
 - b) **Layer 2 — Data Link Layer**
 - c) Layer 3 — Network Layer
 - d) Layer 4 — Transport Layer
-

23. What does MAC stand for in networking?

- a) Main Access Code
 - b) Multiple Access Control
 - c) **Media Access Control**
 - d) Modular Address Controller
-

24. What is a MAC address?

- a) A temporary address assigned by DHCP
 - b) An address used only on the internet
 - c) **A unique, permanent hardware ID assigned to every network device**
 - d) An address used to identify websites
-

25. When a data packet arrives at a switch, the switch:

- a) Broadcasts it to all connected devices
 - b) Sends it to the router for processing
 - c) Drops it if the destination is unknown
 - d) **Reads the destination MAC address and forwards it only to the correct device**
-

26. Which networking device connects different networks and directs data packets between them?

- a) Switch
 - b) Access Point
 - c) Hub
 - d) **Router**
-

27. A router uses which type of address to route data between networks?

- a) MAC address
 - b) Physical address
 - c) **IP address**
 - d) Port address
-

28. Which tool does a router use to find the best path for data packets?

- a) MAC table
 - b) DNS table
 - c) **Routing table**
 - d) DHCP table
-

29. An Access Point (AP) is used to:

- a) Connect two different networks using IP addresses
 - b) Assign IP addresses to devices on a network
 - c) **Connect wireless devices to a wired network**
 - d) Translate domain names to IP addresses
-

30. What is the first thing a switch does when it receives a data packet for the first time?

- a) Drops the packet
 - b) Sends it to the router
 - c) Reads the IP address
 - d) **Broadcasts it to all connected devices until it learns the destination address**
-

Topic: Network Topologies

31. What is a network topology?

- a) The speed at which data travels through a network
 - b) The type of cables used in a network
 - c) **The arrangement of devices and connections in a computer network**
 - d) The security method used to protect a network
-

32. In which topology do all devices share a single communication line?

- a) Star
 - b) Ring
 - c) Mesh
 - d) **Bus**
-

33. What happens in a Bus topology if the main cable fails?

- a) Only the nearest devices go offline
 - b) The router takes over the connection
 - c) Data is rerouted automatically
 - d) **The entire network goes down**
-

34. In which topology are all devices connected to a central switch or hub?

- a) Bus
 - b) Ring
 - c) Mesh
 - d) **Star**
-

35. In a Star topology, what happens if the central switch or hub fails?

- a) Only the first connected device goes offline
 - b) Data is rerouted through other devices
 - c) The network continues to work normally
 - d) **The entire network goes down**
-

36. Which topology is compared to a relay race where a baton passes from one runner to the next in a circle?

- a) Bus
 - b) Star
 - c) **Ring**
 - d) Mesh
-

37. In a Ring topology, in which direction does data travel?

- a) Both directions simultaneously
 - b) In random directions
 - c) From hub to all devices
 - d) **In one direction only, passing through each device**
-

38. Which network topology provides the highest redundancy and reliability?

- a) Bus
 - b) Star
 - c) Ring
 - d) **Mesh**
-

39. In a Mesh topology, each device is connected to:

- a) Only the central hub
 - b) Only its two nearest neighbors
 - c) The main bus cable
 - d) **Every other device in the network**
-

40. Which topology is most suitable when network reliability is the highest priority?

- a) Bus
 - b) Star
 - c) Ring
 - d) **Mesh**
-

41. A school principal's office connected to all classrooms through intercoms is an analogy for which topology?

- a) Bus
 - b) Ring
 - c) Mesh
 - d) **Star**
-

42. Which topology is the easiest and cheapest to set up but the least reliable?

- a) Star
 - b) Ring
 - c) Mesh
 - d) **Bus**
-

Topic: Transmission Modes

43. In which transmission mode does data flow in only one direction?

- a) Full-Duplex
 - b) Half-Duplex
 - c) **Simplex**
 - d) Duplex
-

44. Which of the following is an example of Simplex communication?

- a) A telephone conversation
 - b) A walkie-talkie exchange
 - c) A video call
 - d) **A keyboard sending data to a computer**
-

45. In Half-Duplex communication, data:

- a) Flows only in one direction
 - b) Flows in both directions at the same time
 - c) **Flows in both directions, but not simultaneously**
 - d) Does not travel through a medium
-

46. Which of the following is the best real-life example of Half-Duplex communication?

- a) A telephone call
 - b) A keyboard connected to a computer
 - c) **A walkie-talkie conversation**
 - d) A television broadcast
-

47. In Full-Duplex communication, both devices can:

- a) Only receive data at the same time
 - b) Only send data at the same time
 - c) Send and receive data, but not at the same time
 - d) **Send and receive data simultaneously**
-

48. Which of the following is the best example of Full-Duplex communication?

- a) A keyboard sending input to a computer
 - b) A walkie-talkie
 - c) A television remote control
 - d) **A telephone conversation**
-

49. Which transmission mode allows the highest efficiency of communication?

- a) Simplex
 - b) Half-Duplex
 - c) **Full-Duplex**
 - d) Unidirectional
-

50. The first telephones used which mode of communication?

- a) Full-Duplex
 - b) Simplex
 - c) **Half-Duplex**
 - d) Mesh
-

Topic: OSI Model

51. What does OSI stand for?

- a) Operating System Interface
 - b) Online Security Infrastructure
 - c) **Open Systems Interconnection**
 - d) Optical Signal Integration
-

52. How many layers does the OSI model have?

- a) 4
 - b) 5
 - c) 6
 - d) **7**
-

53. Which layer of the OSI model is responsible for the physical transmission of raw data bits through a medium?

- a) Data Link Layer
 - b) Network Layer
 - c) Transport Layer
 - d) **Physical Layer**
-

54. Which layer of the OSI model handles node-to-node data transfer and error detection?

- a) Physical Layer
 - b) **Data Link Layer**
 - c) Network Layer
 - d) Transport Layer
-

55. Which layer of the OSI model determines the best path for data to travel from source to destination?

- a) Physical Layer
 - b) Data Link Layer
 - c) **Network Layer**
 - d) Transport Layer
-

56. The Network Layer uses which type of address to route data?

- a) MAC address
 - b) Port number
 - c) **IP address**
 - d) Domain name
-

57. Which layer of the OSI model ensures reliable end-to-end data transfer?

- a) Network Layer
 - b) Session Layer
 - c) **Transport Layer**
 - d) Application Layer
-

58. Which protocol does the Transport Layer use to ensure reliable data transfer?

- a) HTTP
 - b) FTP
 - c) **TCP**
 - d) DNS
-

59. Which layer of the OSI model manages the establishment, maintenance, and termination of sessions between applications?

- a) Transport Layer
 - b) **Session Layer**
 - c) Presentation Layer
 - d) Application Layer
-

60. Which layer of the OSI model translates data between the application layer and the network, and handles encryption?

- a) Session Layer
 - b) **Presentation Layer**
 - c) Application Layer
 - d) Transport Layer
-

61. Which layer of the OSI model is closest to the end user and provides network services directly to applications?

- a) Session Layer
 - b) Presentation Layer
 - c) Transport Layer
 - d) **Application Layer**
-

62. Web browsing and email are examples of services provided by which OSI layer?

- a) Transport Layer
 - b) Network Layer
 - c) Session Layer
 - d) **Application Layer**
-

63. The analogy of a GPS system finding the best route from home to school represents which OSI layer?

- a) Physical Layer
- b) Transport Layer
- c) **Network Layer**

- d) Session Layer
-

64. The analogy of a translator converting a book from one language to another represents which OSI layer?

- a) Session Layer
 - b) Application Layer
 - c) **Presentation Layer**
 - d) Transport Layer
-

65. Which OSI layer includes physical components like cables, hubs, and connectors?

- a) Data Link Layer
 - b) Network Layer
 - c) **Physical Layer**
 - d) Transport Layer
-

Topic: IPv4 and IPv6

66. IPv4 uses a how many bit address scheme?

- a) 16-bit
 - b) **32-bit**
 - c) 64-bit
 - d) 128-bit
-

67. How many unique addresses can IPv4 support?

- a) About 1 billion
 - b) About 2 billion
 - c) **Approximately 4.3 billion**
 - d) Approximately 340 undecillion
-

68. Which of the following is a valid IPv4 address?

- a) 2001:0000:130F:0000
- b) **192.168.1.1**

- c) FF:AA:BB:CC:DD:EE
 - d) www.example.com
-

69. IPv4 addresses are written in:

- a) Eight groups of hexadecimal numbers
 - b) Binary notation with 32 characters
 - c) **Four sets of decimal numbers, each ranging from 0 to 255**
 - d) Alphabetic codes separated by colons
-

70. IPv6 uses how many bits for its address scheme?

- a) 32-bit
 - b) 64-bit
 - c) 96-bit
 - d) **128-bit**
-

71. Why was IPv6 developed?

- a) To make internet connections faster
 - b) To replace TCP/IP
 - c) **To address the depletion of IPv4 addresses due to the rapid growth of the internet**
 - d) To improve network security
-

72. Which of the following is a valid IPv6 address?

- a) 192.168.0.1
 - b) FF:EE:DD:CC:BB:AA
 - c) **2001:0000:130F:0000:0000:0900:876A:130B**
 - d) 10.0.0.255
-

73. IPv6 can provide approximately how many unique addresses?

- a) 4.3 billion
- b) 1 trillion
- c) 1 quadrillion

- d) **340 undecillion (an almost limitless number)**
-

Topic: Network Protocols — DNS, DHCP, TCP/IP

74. What does DNS stand for?

- a) Data Network System
 - b) Digital Name Service
 - c) **Domain Name System**
 - d) Dynamic Node Server
-

75. What is the primary function of DNS?

- a) To assign IP addresses automatically to devices
 - b) To encrypt data on the network
 - c) To monitor network traffic
 - d) **To translate domain names into IP addresses**
-

76. When you type www.example.com into a browser, which service translates it into an IP address?

- a) DHCP
 - b) HTTP
 - c) **DNS**
 - d) TCP
-

77. What does DHCP stand for?

- a) Data Host Control Protocol
 - b) **Dynamic Host Configuration Protocol**
 - c) Digital Hardware Configuration Program
 - d) Domain Host Communication Protocol
-

78. What is the primary function of DHCP?

- a) To translate domain names to IP addresses
- b) To encrypt data on the network

- c) **To automatically assign IP addresses to devices on a network**
 - d) To monitor network traffic
-

79. When a device connects to a Wi-Fi network and receives an IP address automatically, which protocol is responsible?

- a) DNS
 - b) TCP
 - c) HTTP
 - d) **DHCP**
-

80. What does TCP/IP stand for?

- a) Terminal Control Protocol / Internet Process
 - b) **Transmission Control Protocol / Internet Protocol**
 - c) Transfer Communication Protocol / Intranet Protocol
 - d) Technical Computing Protocol / Internet Program
-

81. Which protocol in the TCP/IP suite ensures reliable data transfer?

- a) IP
 - b) UDP
 - c) HTTP
 - d) **TCP**
-

82. Which protocol in the TCP/IP suite handles addressing and routing of data packets?

- a) TCP
 - b) UDP
 - c) **IP**
 - d) FTP
-

83. UDP (User Datagram Protocol) is characterized by:

- a) Very slow but highly reliable data transfer
- b) Two-way handshake before data transfer

- c) **Faster but less reliable data transfer compared to TCP**
 - d) Encryption of all data before sending
-

84. Which protocol is used for transferring web pages over the internet?

- a) FTP
 - b) SMTP
 - c) DNS
 - d) **HTTP**
-

85. Which of the following is a correct sequence showing how protocols relate?

- a) HTTP → IP → DNS → TCP
 - b) DNS → UDP → MAC → HTTP
 - c) **HTTP uses TCP, which uses IP, which uses the physical network**
 - d) DHCP → DNS → HTTP → MAC
-

Topic: Network Security

86. Which of the following best describes the purpose of network security?

- a) To increase the speed of data transmission
 - b) To organize data into packets
 - c) To assign IP addresses to devices
 - d) **To protect data and prevent unauthorized access to computer networks**
-

87. Which network security method monitors and controls incoming and outgoing network traffic?

- a) Encryption
 - b) Antivirus software
 - c) **Firewall**
 - d) DHCP
-

88. What does encryption do to data?

- a) Deletes unauthorized data
- b) Sends data faster

- c) Assigns IP addresses to data packets
 - d) **Converts data into a secure format so only authorized users can read it**
-

89. Which network security method detects and removes malicious software?

- a) Firewall
 - b) Encryption
 - c) **Antivirus software**
 - d) DNS
-

90. Which of the following is a reason why network security is important?

- a) It increases the number of IP addresses available
 - b) It makes data transfer faster
 - c) **It protects sensitive information from unauthorized access**
 - d) It connects different networks together
-

91. "Maintaining Privacy" as a network security goal means:

- a) Ensuring data travels at maximum speed
 - b) Preventing overheating of network devices
 - c) Keeping IP addresses hidden from routers
 - d) **Safeguarding personal and confidential information from being compromised**
-

92. "Ensuring Availability" as a network security goal means:

- a) Backing up data on cloud servers
 - b) Encrypting all data before transmission
 - c) Blocking all incoming network traffic
 - d) **Ensuring that network resources are accessible to authorized users at all times**
-

93. A combination of which three tools provides the most robust network security?

- a) DNS, DHCP, and TCP
- b) Router, Switch, and Access Point
- c) **Firewall, Encryption, and Antivirus Software**

- d) Hub, Modem, and Repeater
-

Topic: Types of Networks

94. Which type of network is used for communication between personal devices within a few metres?

- a) LAN
 - b) MAN
 - c) WAN
 - d) **PAN**
-

95. A Bluetooth connection between a smartphone and a wireless headset is an example of which network?

- a) LAN
 - b) WAN
 - c) CAN
 - d) **PAN**
-

96. Which type of network connects computers and devices within a limited area such as a school or office building?

- a) PAN
 - b) MAN
 - c) WAN
 - d) **LAN**
-

97. The computer network in a school lab connecting all computers is an example of:

- a) PAN
 - b) **LAN**
 - c) MAN
 - d) WAN
-

98. Which type of network spans a city or large campus, connecting multiple LANs?

- a) PAN
 - b) LAN
 - c) **MAN**
 - d) WAN
-

99. A MAN can cover an area of up to:

- a) 5 kilometres
 - b) 10 kilometres
 - c) 25 kilometres
 - d) **50 kilometres**
-

100. Which type of network covers the largest geographical area, connecting multiple LANs and MANs across countries?

- a) PAN
 - b) LAN
 - c) MAN
 - d) **WAN**
-

101. The Internet is the largest example of which type of network?

- a) LAN
 - b) MAN
 - c) CAN
 - d) **WAN**
-

102. Which type of network connects multiple LANs within a university campus or business park?

- a) PAN
 - b) LAN
 - c) MAN
 - d) **CAN**
-

103. An Intranet is best described as:

- a) A public network accessible to everyone on the internet
 - b) A type of wireless network using Bluetooth
 - c) **A private network inside an organization that uses internet technology**
 - d) A network that spans multiple cities
-

104. Which type of network would a multinational company use to connect its offices in different countries?

- a) PAN
 - b) LAN
 - c) CAN
 - d) **WAN**
-

Topic: Real-World Applications

105. Which technology allows doctors to consult with patients remotely through video calls and shared medical images?

- a) LMS
 - b) EHR
 - c) DNS
 - d) **Telemedicine**
-

106. What does EHR stand for in healthcare networks?

- a) **Electronic Health Records**
 - b) Encrypted Hospital Records
 - c) External Health Repository
 - d) Electronic Hub Router
-

107. Which of the following is an example of a Learning Management System (LMS)?

- a) Facebook
- b) WhatsApp
- c) Instagram
- d) **Google Classroom**

108. Which of the following LMS platforms is commonly used in Pakistani universities?

- a) Canvas
 - b) Blackboard
 - c) **Moodle**
 - d) Google Classroom
-

109. What is "flipped learning"?

- a) A teaching method where students teach the teacher
 - b) A method of learning where students study backward from the exam
 - c) **A method where students access lecture material online before class, then discuss it in class**
 - d) A method where all learning is done in the classroom without technology
-

110. Computer networks benefit healthcare by:

- a) Making hospital buildings larger
 - b) Replacing doctors with computers
 - c) Increasing the number of hospital beds
 - d) **Enabling faster access to patient records and reducing medical errors**
-

111. During the COVID-19 pandemic, researchers used networks to:

- a) Build physical laboratories in multiple countries
 - b) Send physical samples between hospitals
 - c) **Share genomic virus data internationally in real time to speed up vaccine development**
 - d) Increase internet connection speed globally
-

112. Which of the following is a benefit of networks in education?

- a) Reducing the number of teachers needed
- b) Removing the need for physical schools
- c) **Providing access to knowledge regardless of geographic location**
- d) Preventing students from accessing incorrect information

113. A company uses a private internal website to share information only with its employees. This is called:

- a) Extranet
 - b) WAN
 - c) **Intranet**
 - d) CAN
-

114. Which of the following correctly matches a network type with its real-world example?

- a) PAN — University campus network
 - b) LAN — International company connecting offices across countries
 - c) MAN — Bluetooth headset connected to a phone
 - d) **LAN — School computer lab network**
-

115. Packet switching in a router is compared to which real-life scenario in the chapter?

- a) A relay race on a circular track
 - b) A GPS finding the shortest route
 - c) **Passengers split into groups and assigned different flights to reach the same destination**
 - d) A phone call session being established and terminated
-
-

Section B: Short Questions

Answer each question in 2–5 sentences. No answers are provided — use your notes and textbook.

Topic: Network as a System

1. Define a computer network.
2. What is meant by the term "network of networks"? What is it referring to?
3. List the four primary components of a computer network and briefly describe each.
4. What is a node in a computer network? Give two examples.

5. What is a link in a computer network? Give one example of a wired link and one of a wireless link.
 6. State three main objectives of a computer network.
 7. What is resource sharing in the context of computer networks? Give one example.
 8. How do computer networks enable data communication? Give one real-life example.
 9. What is meant by connectivity and collaboration in computer networks?
-

Topic: Data Communication

1. Define data communication.
 2. List the five key components of data communication.
 3. What is the role of the sender in data communication? Give an example.
 4. What is the role of the receiver in data communication? Give an example.
 5. What is a message in the context of data communication? What forms can it take?
 6. What is a protocol in data communication? Why is it important?
 7. Give two examples of communication mediums used in data communication.
 8. Differentiate between a wired medium and a wireless medium in data communication.
-

Topic: Networking Devices

1. What is a switch? What is its role in a network?
 2. At which OSI layer does a switch operate? What address does it use to forward data?
 3. What is a MAC address? What makes it unique?
 4. Explain how a switch forwards data when it receives a packet. What does it do the first time and after it learns the addresses?
 5. What is a router? How does it differ from a switch?
 6. Write the role of routers in a computer network.
 7. What is a routing table? How does a router use it?
 8. What is an Access Point (AP)? What problem does it solve?
 9. How does an Access Point work? What does it do with data from the wired network?
 10. Differentiate between a switch and a router.
-

Topic: Network Topologies

1. What is a network topology?
2. Define Bus topology and state one advantage and one disadvantage.

3. Define Star topology. What is the role of the central hub or switch?
 4. What happens in a Star topology if the central switch fails?
 5. Define Ring topology. In which direction does data travel?
 6. What is Mesh topology? Why is it considered the most reliable?
 7. Differentiate between Bus topology and Star topology.
 8. Differentiate between Ring topology and Mesh topology.
 9. Which topology would you recommend for a hospital where network reliability is critical? Justify your answer.
-

Topic: Transmission Modes

1. What are transmission modes in computer networks? Name all three types.
 2. Define Simplex communication. Give one example.
 3. Define Half-Duplex communication. Give one example.
 4. Define Full-Duplex communication. Give one example.
 5. What is the difference between Half-Duplex and Full-Duplex communication?
 6. Differentiate between Simplex and Full-Duplex communication.
 7. Which transmission mode allows the most efficient two-way communication? Why?
-

Topic: OSI Model

1. What does OSI stand for? What is the purpose of the OSI model?
 2. How many layers does the OSI model have? List them in order from Layer 1 to Layer 7.
 3. What is the function of the Physical Layer (Layer 1) in the OSI model?
 4. What is the function of the Data Link Layer (Layer 2)? Give an example analogy.
 5. What is the function of the Network Layer (Layer 3)? What address does it use?
 6. What is the function of the Transport Layer (Layer 4)? Which protocol does it use?
 7. What is the function of the Session Layer (Layer 5)? Give an example.
 8. What is the function of the Presentation Layer (Layer 6)?
 9. What is the function of the Application Layer (Layer 7)? Give two examples of services it provides.
 10. Define any three layers of the OSI model and explain their functions.
-

Topic: IPv4 and IPv6

1. What is an IP address? What is its purpose?

2. What is IPv4? How many bits does it use?
 3. How many unique addresses can IPv4 support? Write the formula used to calculate this.
 4. Write an example of an IPv4 address and explain its format.
 5. What is IPv6? Why was it developed?
 6. How many bits does IPv6 use for its address scheme?
 7. Write an example of an IPv6 address.
 8. Differentiate between IPv4 and IPv6.
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Topic: Protocols — DNS, DHCP, TCP/IP

1. What is a network protocol? Give two examples.
 2. What is DNS? What problem does it solve?
 3. Explain how DNS works when a user types a website address into a browser.
 4. What is DHCP? What is its purpose?
 5. Explain how DHCP works when a device connects to a Wi-Fi network.
 6. What does TCP/IP stand for? What is it used for?
 7. What is the role of TCP (Transmission Control Protocol) in the TCP/IP suite?
 8. What is the role of IP (Internet Protocol) in the TCP/IP suite?
 9. What is UDP? How does it differ from TCP?
 10. Differentiate between TCP and UDP.
 11. Name four common network protocols and briefly state what each one does.
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Topic: Network Security

1. What is network security? Why is it important?
 2. List four reasons why network security is important.
 3. What is a firewall? How does it contribute to network security?
 4. What is encryption in network security? How does it protect data?
 5. What is antivirus software? What role does it play in network security?
 6. Explain what is meant by "data protection" as a network security goal.
 7. Explain what is meant by "preventing attacks" as a network security goal.
 8. How do firewalls, encryption, and antivirus software work together to provide robust security?
-

Topic: Types of Networks

1. What is a PAN? Give one real-world example.
 2. What is a LAN? Give one real-world example.
 3. What is a MAN? What area does it typically cover?
 4. What is a WAN? Give one real-world example.
 5. What is a CAN? How is it different from a LAN?
 6. Differentiate between LAN and WAN.
 7. Differentiate between MAN and CAN.
 8. What is an Intranet? How is it different from the Internet?
 9. Which type of network would be most suitable for connecting the computers in a single school building? Justify your answer.
 10. Arrange the network types PAN, LAN, CAN, MAN, WAN in order from smallest to largest coverage area.
-

Topic: Real-World Applications

1. How do computer networks benefit businesses? Give two specific examples.
 2. What is an LMS (Learning Management System)? Give two examples of LMS platforms.
 3. How do networks enable virtual classrooms?
 4. What is "flipped learning"? How do networks make it possible?
 5. What are Electronic Health Records (EHR)? List two benefits of EHR systems.
 6. What is telemedicine? Give one real-world example of how it helps patients.
 7. How did networks help scientists respond to the COVID-19 pandemic?
 8. State two benefits of computer networks in education.
 9. State two benefits of computer networks in healthcare.
 10. What is an Intranet? How do companies use it?
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Section C: Long Questions

Answer each question in detail. Use diagrams where applicable. No answers are provided.

1. Discuss the objectives of computer networks in detail. For each objective, provide a real-world example and explain how it benefits individuals or organizations.
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2. Explain the five components of data communication in detail. For each component, provide a clear definition and a real-world example. How do all five components work together to enable successful communication?

3. Describe the four main networking devices: Node, Link, Switch, and Router. Explain how each one works, what address or mechanism it uses, and give a real-world example for each. Clearly differentiate between a switch and a router.

4. What is a network topology? Explain all four network topologies — Bus, Star, Ring, and Mesh — with the following for each:

- Definition
 - How devices are connected
 - Advantages
 - Disadvantages
 - A real-life analogy or example
 - A labeled diagram
-

5. Explain the three modes of data transmission — Simplex, Half-Duplex, and Full-Duplex — with examples, diagrams, and real-life analogies. Compare all three modes and state which is most efficient and why.

6. Define the OSI networking model. Explain all seven layers of the OSI model. For each layer, state:

- Its layer number and name
 - Its primary function
 - A real-life analogy or example
-

7. Describe any three layers of the OSI model in detail. Explain their functions, the protocols they use, and give a real-world analogy for each.

8. Compare and contrast IPv4 and IPv6. In your answer:

- Define both versions
- State the number of bits used by each
- Calculate or state the number of unique addresses each supports

- Provide an example address for each
 - Explain why IPv6 was developed
 - State the key differences between the two
-

9. What are network protocols? Explain the following protocols in detail with examples of how each one works in a real-world scenario:

- HTTP
 - DNS
 - DHCP
 - TCP
 - IP
 - UDP
-

10. Explain the importance of network security. Describe the following three security methods in detail — including how each one works, its advantages, and any limitations:

- Firewall
- Encryption
- Antivirus Software

How do these three methods work together to provide comprehensive network security?

11. Explain the five types of computer networks — PAN, LAN, CAN, MAN, and WAN. For each network type:

- Provide a definition
- State the typical range or area it covers
- Give one real-world example
- State one advantage

Draw a simple diagram for any two of the network types.

12. Describe real-world applications of computer networks in **business** and **education**. For each sector:

- Explain at least two specific ways networks are used
 - Give concrete real-world examples
 - State at least two benefits that networks bring to that sector
-

13. Describe real-world applications of computer networks in **healthcare**. In your answer:

- Explain what Electronic Health Records (EHR) are and how they benefit patients and doctors
 - Explain what telemedicine is and give an example of how it helps patients in remote areas
 - Explain how medical research networks accelerated the response to the COVID-19 pandemic
 - State at least three overall benefits of networks in healthcare
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14. In a Simplex communication system, data is transmitted at a rate of 500 bits per second (bps). Calculate the time required to transmit:

- a) A message of 10 kilobits
- b) A message of 10 kilobytes

Show all working and state your final answers with the correct units.

15. Write a comprehensive essay on the role of computer networks in modern society. In your essay, discuss:

- What a computer network is and why it is important
 - How data communication works
 - The types of networks and where each is used
 - How networks are secured
 - The real-world impact of networks on business, education, and healthcare
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End of Unit 3: Computer Networks — Complete Question Bank